Logical Volume Manager (LVM)

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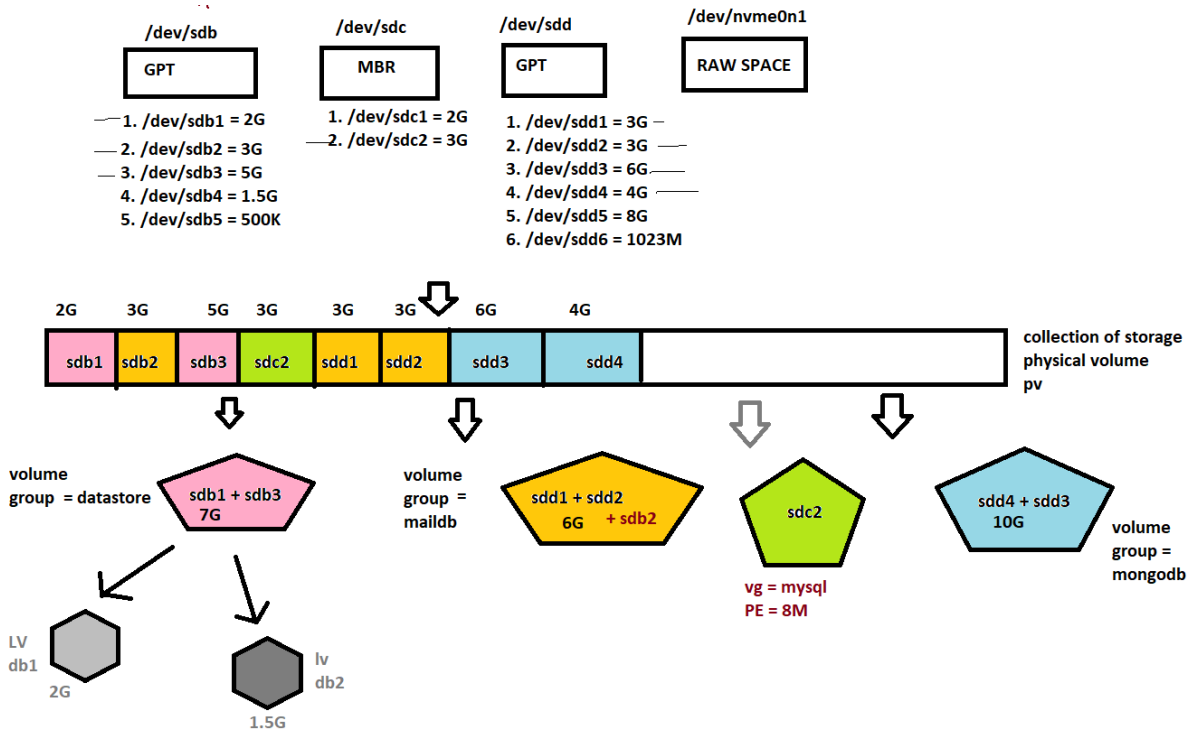

1. **Create partitions**
    a. For MBR partition style : fdisk
    b. For GPT partition style : gdisk
2. Change the partition type using **“t”** option
3. Hex code for lvm is

   a. **MBR = 8e**
   b. **GPT = 8e00**

4. Save the partition table => # udevadm settle
5. **Creating pv :-**

Command ⇒ # pvcreate <partition names with space>
Example ⇒ # pvcreate /dev/sdb1 /dev/sdb2 /dev/sdb3 /dev/sdc2 /dev/sdd1 /dev/sdd2 /dev/sdd3 /dev/sdd4 /dev/sde

6. **Check pv is created or not**
   a. # pvs : for glimpse o/p
   b. # pvdisplay : for detailed o/p

Note : we can create only one pv per system

7. **Creating vg from pv :-**

```
# vgcreate datastore /dev/sdb1 /dev/sdb3
# vgcreate maildb /dev/sdd1 /dev/sdd2
# vgcreate mongodb /dev/sdd3 /dev/sdd4
```

8. **To check vg :-**
   a. # vgs
   b. # vgdisplay

Note :default block size of PE(Physical extend =4M)

9. **To create lvm / lv :-**
    a. **Using PE Count** :-
       # lvcreate -l 512 -n db1 datastore
    b. **Using Size** :-
       # lvcreate -L 1500M -n db2 datastore

10. **To check lvm /lv :**
    a. # lvs
    b. # lvdisplay

11. **Formatting a lv using ext4 / xfs** :-
    a. # mkfs.ext4 /dev/datastore/db1
    b. # mkfs.xfs /dev/datastore/db2

12. **Creating Permanent mount** :-
    a. Open /etc/fstab file
    b. /dev/vgname/lvname mountpoint fs   defaults 0 0
    c. Save and exit

d. # mount -a

13. **Check the disks with mounts**

    # df -Th

14. **Vgcreate with custom PE Size (8M)**

    # vgcreate -s 8M mysql /dev/sdc2

15. **Lvm extending :-**

    a. increase the size of db1 lvm to 2.7G

        1. Check the size of lvm
        2. Check how many blocks required to become 2.7G
        3. # lvextend -l 175 /dev/datastore/db1

    b. increase the size of db2 lvm by 1G

        # lvextend -l +256 /dev/datastore/db2

16. **Updating the new size :-**

    a. if the lvm is formatted by xfs file system :- # xfs_growfs /dev/datastore/db2

b. If the lvm is formatted by ext4 file system :- # resize2fs /dev/datastore/db1

17. Extending pv size :-

```
[root@localhost ~]# pvcreate /dev/sdd1
  Physical volume "/dev/sdd1" successfully created.
[root@localhost ~]#
[root@localhost ~]#
[root@localhost ~]# pvs
  PV         VG        Fmt  Attr PSize   PFree
  /dev/sda2  rhel_vbox lvm2 a--  <19.00g    0
  /dev/sdb1  vg2       lvm2 a--    4.99g 3.92g
  /dev/sdc2  vg1       lvm2 a--  <10.00g 6.31g
  /dev/sdd1            lvm2 ---    2.00g 2.00g
[root@localhost ~]#
```

18. **Extending vg size :-**

```
[root@localhost ~]# vgextend vg1 /dev/sdd1
  Volume group "vg1" successfully extended
[root@localhost ~]#
[root@localhost ~]# vgs
  VG        #PV #LV #SN Attr   VSize   VFree
  rhel_vbox   1   2   0 wz--n- <19.00g     0
  vg1         2   2   0 wz--n-  11.99g <8.31g
  vg2         1   2   0 wz--n-   4.99g  3.92g
[root@localhost ~]#
```

Reducing vg size

```
[root@localhost ~]# vgreduce VolumeGroupName /dev/sdb1
```

19. Reducing lvm size :-

Reduce of lvm size is only possible if the lv is formatted using ext4 file system.

1. Unmount the lvm :
    a. Temp : umount /dev/vg2/lv3 /mnt/lv3
    b. Perm : delete the fstab file entry regarding this lvm
2. Start file system checking :

```
[root@localhost ~]# e2fsck -f /dev/vg2/lv3
```

3. Now resize the lvm :

```
[root@localhost ~]# lvresize -rv -L 100M /dev/vg2/lv3
```

4. Now perform fstab entry
5. mount -a

Removing lv :-

1. Unmount the mount point ⇒

```
[root@localhost ~]# umount /dev/vg1/lv2 /mnt/lv2
umount: /mnt/lv2: not mounted.
```

2. Remove all lvm entries from /etc/fstab file and update using mount -a
3. remove lv :-

```
[root@localhost ~]# lvremove /dev/vg1/lv1
Do you really want to remove active origin logical volume vg1/lv1 with 1 snapshots(s)? [y/n]: y
  Logical volume "snap1" successfully removed.
  Logical volume "lv1" successfully removed.
[root@localhost ~]#
[root@localhost ~]# lvremove /dev/vg1/lv2
Do you really want to remove active logical volume vg1/lv2? [y/n]: y
  Logical volume "lv2" successfully removed.
[root@localhost ~]#
[root@localhost ~]# lvremove /dev/vg2/lv3
Do you really want to remove active logical volume vg2/lv3? [y/n]: y
  Logical volume "lv3" successfully removed.
[root@localhost ~]#
[root@localhost ~]# lvremove /dev/vg2/lv4
Do you really want to remove active logical volume vg2/lv4? [y/n]: y
  Logical volume "lv4" successfully removed.
```

20. **Removing vg :-**

```
[root@localhost ~]# vgremove vg1
  Volume group "vg1" successfully removed
[root@localhost ~]#
[root@localhost ~]# vgremove vg2
  Volume group "vg2" successfully removed
[root@localhost ~]#
```

21. **Removing partition from pv :-**

```
[root@localhost ~]# pvremove /dev/sdb1 /dev/sdc2 /dev/sdd1
  Labels on physical volume "/dev/sdb1" successfully wiped.
  Labels on physical volume "/dev/sdc2" successfully wiped.
  Labels on physical volume "/dev/sdd1" successfully wiped.
[root@localhost ~]#
```